

Investigating “Intelligent Slop”

A Small Case Study on How Reddit Content Affects an LLM

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Abstract

Recent work has raised the concern that training large language models on low-quality data can degrade capabilities (“brain rot”) (Xing et al. 2025). However, “low quality” covers many different kinds of content, from incoherent noise to creative but unconventional writing. This report is a small case study asking a simple question: *if we fine-tune on meme-like content that still has structure, do we see anything interesting happen to reasoning benchmarks?*

Using LoRA fine-tuning on Llama-3.1-8B-Instruct with data from two contrasting Reddit communities, I evaluate model behavior across a suite of benchmarks while varying the amount of fine-tuning data ($n \in \{8, 16, 32, 64\}$). The results are mixed. Some tasks improve (most notably GSM8K under r/fifthworldproblems data), many degrade, and there are noticeable shifts in personality-style evaluations. Overall, this is not a decisive win for any single story, but it is a useful snapshot of how “internet-y” data can push a model in multiple directions at once.

1 Introduction and Motivation

A growing concern is that the internet is filling with low-effort content and synthetic text, which may affect the quality of future LLM training data. The “brain rot” paper studied this explicitly and showed that training on low-quality social media data can degrade downstream performance (Xing et al. 2025). Their setup used Twitter/X data and two quality proxies, one based on engagement and one based on semantic features.

My starting point was a practical observation: engagement is not the same thing as quality. A short viral post can be popular without being informative, and niche technical discussion can be high quality while receiving low engagement. That is not a fatal flaw, but it suggests a follow-up question.

1.1 A Small Follow-Up Question

What if some content that looks like “slop” is still cognitively demanding?

Absurdist humor communities often produce writing that is ridiculous on the surface but still involves wordplay, constraints, and nontrivial inference. The goal here is not to claim that this kind of data is “good” in general. It is to treat it as a concrete corner of the internet and see what it does to a model when used for fine-tuning.

1.2 The “Intelligent Slop” Idea (as a Case Study)

I use “intelligent slop” as a label for content that is unconventional or absurdist but still structured enough to require reasoning to produce and interpret. The working expectation was simple:

- This data will likely come with tradeoffs and harm some capabilities.

- It might still produce localized improvements on certain reasoning-style tasks.

This report is written in that spirit. It is a case study of memes and internet writing, not a definitive verdict.

2 Methodology

2.1 Data Collection

I collected data from two contrasting Reddit communities that represent different styles of non-standard content.

2.1.1 r/fifthworldproblems (Absurdist / Creative)

This community features absurdist “problems” and “solutions” that often involve:

- Wordplay and puns (for example “cubed up” versus “squared up”)
- Logical paradoxes (for example “all roads lead to Rome, how do I leave?”)
- Abstract reasoning about impossible scenarios
- Creative problem-solving under unusual constraints

Example entries:

Q: “My doctor gave me a year to live. So I killed him and the judge gave me 20 years. How else can I prolong my existence?”

A: “If you commit treason, you’ll receive a life sentence if found guilty.”

Q: “I decided to visit Rome, but all roads lead to Rome, so when I try to leave I end up in Rome again.”

A: “Go up.”

2.1.2 r/benignexistence (Mundane / Wholesome)

As a comparison condition, I collected data from a community centered on everyday, mundane observations:

- Slice-of-life descriptions of daily routines
- Wholesome interactions involving family, pets, or neighbors
- Minimal creative or abstract reasoning, focusing on ordinary experiences

Example entry:

“My cat has a bedtime routine. Every night when we start getting ready for bed, he drinks water, uses his litter box, then comes to lay on my pillow...”

2.1.3 Data Filtering

Raw Reddit data includes noise such as meta-comments, reactions, and off-topic replies. For the r/fifthworldproblems dataset, I used GPT-4o filtering and kept pairs that:

- Present a clear “problem” in the post
- Provide an actual “solution attempt” in the response (not only reactions)
- Have output length between 6–50 words

Final dataset sizes:

- r/fifthworldproblems: 291 examples after filtering
- r/benignexistence: 300 examples

2.2 Training Setup

Table 1: Training configuration.

Parameter	Value
Base Model	Llama-3.1-8B-Instruct
Fine-tuning Method	LoRA (Low-Rank Adaptation)
LoRA Rank (r)	16
LoRA Alpha (α)	32
LoRA Dropout	0.05
Target Modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
Optimizer	AdamW
Learning Rate	2e-4
LR Schedule	Cosine decay
Epochs	10
Batch Size	2 per device
Max Sequence Length	512 tokens
Hardware	2× NVIDIA RTX A6000 (48GB each)

To probe data-size effects, I trained models with $n \in \{8, 16, 32, 64\}$ examples for both datasets.

2.3 Evaluation Benchmarks

I evaluated models on a set of benchmarks spanning math, commonsense, logic, factuality, and a personality-style trait benchmark. Evaluations used the `lm_eval` harness with consistent settings (3-shot for most tasks, 250-sample cap for expensive benchmarks).

Table 2: Representative evaluation benchmarks.

Category	Benchmark	Description
Math Reasoning	GSM8K	Grade school math word problems
Common Sense	HellaSwag	Sentence completion
Riddles	RiddleSense	Riddle comprehension
Factuality	TruthfulQA	Resistance to common misconceptions
Logic (BBH)	Boolean Expressions	Evaluate logical expressions
Personality	TRAIT Benchmark	Big Five and Dark Triad assessment

3 Results

3.1 Overall Snapshot

Table 3 shows a representative subset of benchmark results. Green indicates above baseline and red indicates below baseline.

Table 3: Accuracy (%) across benchmarks for all training configurations. **Green** indicates above baseline, **Red** indicates below baseline.

Benchmark	Base	Fifth World (Absurdist)				Benign (Mundane)			
		n8	n16	n32	n64	n8	n16	n32	n64
GSM8K	65.6	69.2	70.8	69.2	68.4	69.6	51.6	62.4	57.2
HellaSwag	56.4	58.4	57.6	58.0	56.4	56.8	56.0	58.0	56.0
BBH Boolean	90.8	91.2	91.6	90.8	91.2	88.0	90.0	88.4	89.6
TruthfulQA	53.2	50.8	49.7	44.0	44.2	49.4	45.3	47.2	46.8
RiddleSense	81.6	69.4	75.5	71.4	75.5	75.5	69.4	69.4	67.4

3.2 The Fun Part First: GSM8K Improves Under r/fifthworldproblems Data

The most eye-catching result is GSM8K. Fine-tuning on r/fifthworldproblems improves over baseline across all tested n values in Table 3.

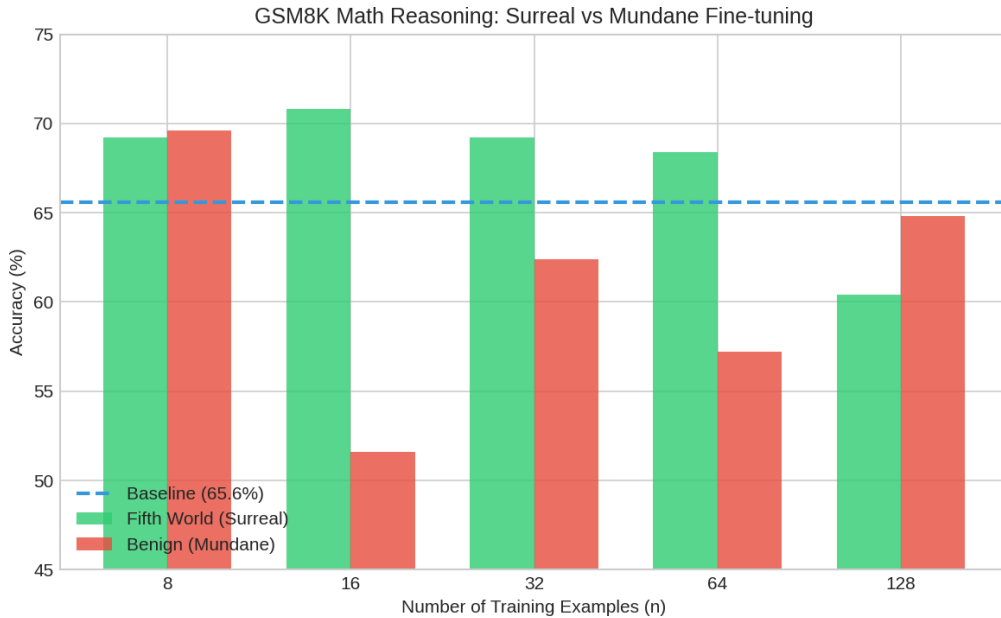


Figure 1: GSM8K performance across training set sizes. r/fifthworldproblems improves over baseline for $n \in \{8, 16, 32, 64\}$, while r/benignexistence is unstable and drops sharply at $n = 16$.

This is not a slam dunk. With a 250-sample cap, the standard error is on the order of a few percentage points, so these gains are suggestive rather than decisive. Still, the consistency across n is what makes it interesting.

If nothing else, it gives the people who are online 24/7 a new line to boast about: “the memes made my model better at math.”

3.3 The Rest: Mixed Tradeoffs, Not a Single Story

Most other benchmarks show degradation for both datasets, and the pattern depends on content type.



Figure 2: Benchmark comparison at $n = 16$. Improvements exist but are not uniform across tasks.

A reasonable summary is:

- There are real-looking improvements on some tasks (GSM8K being the main one here).
- Many tasks still drop, sometimes sharply.
- The two datasets do not fail in the same way, which suggests “what kind of slop” matters.

3.4 Personality Shifts (Expected, and Still Worth Measuring)

I also tracked personality-style trait scores. This is not because “personality” is the goal, but because it is a quick way to see whether the fine-tuning data pushes the model’s tone and tendencies in a measurable direction.

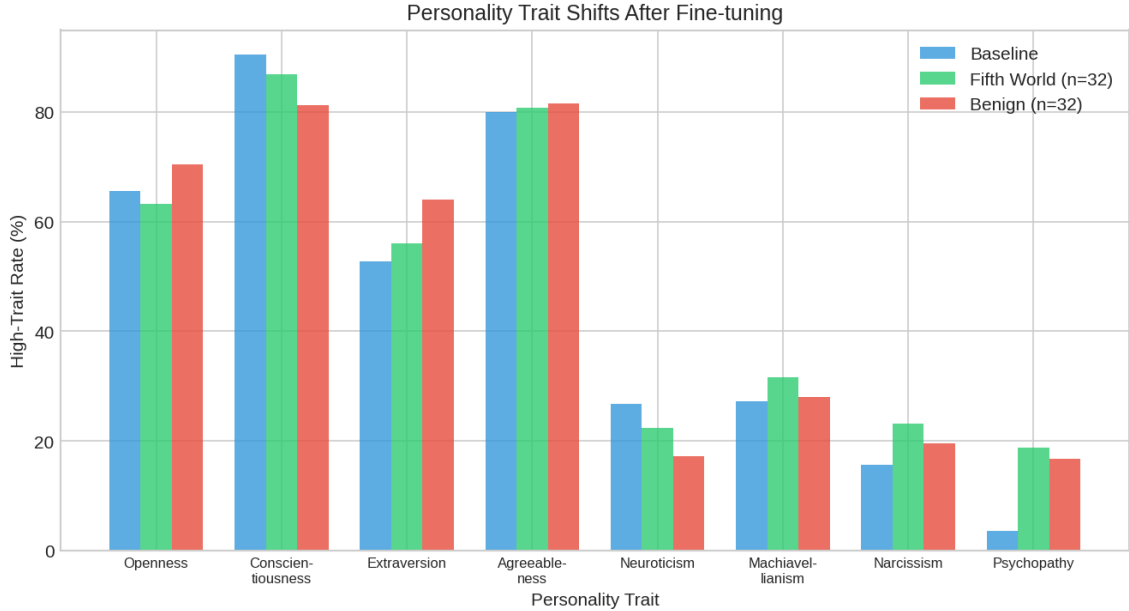


Figure 3: Personality trait shifts after fine-tuning ($n = 32$).

Table 4: Personality trait changes ($n = 32$ vs baseline).

Trait	Baseline	Fifth World Δ	Benign Δ
Openness	65.6%	-2.4 pp	+4.8 pp
Conscientiousness	90.4%	-3.6 pp	-9.2 pp
Extraversion	52.8%	+3.2 pp	+11.2 pp
Psychopathy	3.6%	+15.2 pp	+13.2 pp
Narcissism	15.6%	+7.6 pp	+4.0 pp

Both datasets shift trait scores. In this run, r/fifthworldproblems produced larger increases on some dark-triad-style dimensions. The main point is that internet content does not only move accuracy numbers, it also moves style.

4 Discussion

4.1 So, Did “Intelligent Slop” Help?

The honest answer is: **sometimes** (not really). GSM8K improves consistently under r/fifthworldproblems fine-tuning in this setup, but other benchmarks do not show the same clear uplift. The right interpretation is not “it works” or “it fails,” but “it pushes capabilities around, and the pushes are uneven.”

4.2 A Few Notes on Interpretation

A few practical caveats matter here:

- **Small n regime:** These experiments intentionally stay in the low-data regime. That makes the results noisy, but it also reflects what people do when they fine-tune quickly.
- **Format effects:** Both datasets are informal Reddit writing. Some drops may be caused by a format shift rather than the underlying content.
- **Not an optimization claim:** This is not proposing a new training recipe. It is a case study on what happens when you feed a model a particular slice of the internet.

4.3 What This Case Study Suggests

Even with mixed results, a few things come through:

1. **Memes can be structured:** r/fifthworldproblems is not only noise, and the GSM8K improvements are consistent enough to be worth following up.
2. **Tradeoffs are real:** gains on one axis can coexist with losses elsewhere.

5 Conclusion

This report is a small case study on how two kinds of Reddit data affect an 8B instruct model under LoRA fine-tuning. The headline is that the effects are mixed and content-dependent.

Fine-tuning on r/fifthworldproblems produced consistent GSM8K improvements across $n \in \{8, 16, 32, 64\}$ in this setup, which was not what one would expect from “slop” at first glance. At the same time, other benchmarks dropped, and personality-style trait scores shifted in noticeable ways. If nothing else, this supports a simple lesson: “low quality” is not one thing, and internet data can nudge models in more than one direction.

References

Xing, Shuo et al. (2025). “LLMs Can Get” Brain Rot”!” In: *arXiv preprint arXiv:2510.13928*.